

The empirical recurrence for OEIS A209117, proved from an exact model

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Abstract

OEIS A209117 carries an empirical recurrence of order 10. It is true, and it is decidable rather than empirical. The entry's sequence is given exactly by a Burnside sum over the necklace group, from which a provably satisfies a MONIC linear recurrence of order $S = 46$, derived below and not assumed. The residual of the conjectured recurrence satisfies the same one, so whether it annihilates a is settled by a finite exact computation. No transfer matrix is involved and the count is not a walk.

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1 The conjecture

OEIS A209117 is “Number of 5-bead necklaces labeled with numbers $-n..n$ not allowing reversal, with sum zero and avoiding the patterns $z\ z+1\ z+2$ and $z\ z-1\ z-2$.” It has offset 1 and begins

5, 63, 261, 741, 1683, 3325, 5935, 9835, ...

The entry states, as a conjecture contributed by its author and never marked settled:

Empirical: $a(n) = 3*a(n-1) - 3*a(n-2) + 2*a(n-3) - 2*a(n-4) + 2*a(n-6) - 2*a(n-7) + 3*a(n-8) - 3*a(n-9) + a(n-10)$.

As of the “Last modified” line on the live entry (Jul 08 2018 21:30:59, revision 13) this is still recorded as empirical, and nothing on the entry records it as proved.

2 The count is a Burnside sum

The entry counts 5-bead necklaces labelled from $-n..n$ with sum zero under a further condition on consecutive beads, the beads identified under the cyclic group of rotations, written G . Nothing is a walk here: the bead count is fixed and the ALPHABET grows.

By Burnside's lemma the number of orbits is $\frac{1}{|G|} \sum_{g \in G} |\text{Fix}(g)|$. A labelling fixed by g is constant on the orbits of g , so it is a choice of one value per orbit. For a rotation by d with $c = \gcd(5, d)$ the fixed labellings are exactly the c -periodic words, the whole sum is $(5/c)$ times the sum of one period, and the condition on the 5-periodic extension is the condition on the length- c cyclic word; for a reflection they are the palindromes, determined by $\lceil (5+1)/2 \rceil$ values, and the condition is again local in those. Every term is therefore a count of integer points of

$$\sum_i w_i y_i = 0, \quad -n \leq y_i \leq n,$$

w_i the orbit sizes, under a condition on consecutive beads. The forbidden windows the entry names are defined by EQUALITIES between bead values, with fixed offsets, so inclusion-exclusion over them turns each term into a count under a set of affine identifications — one convolution each — and nothing is enumerated over the alphabet.

3 The bound

Each term is a lattice-point count over a union of relatively open rational cones in the orbit variables, hence a quasi-polynomial in n ; a Burnside average of quasi-polynomials is one. The period is read off the arrangement: every ray of every cell is cut out by as many independent hyperplanes as there are variables, so by Cramer its primitive generator has its n -coordinate dividing the determinant of their y -parts, and a ray leaving the box or the sum-zero plane bounds no cell and is dropped. The multiplicity of each cyclotomic factor is bounded by the number of distinct rays whose height it divides, and by the dimension of the cone. Taking the maximum over g of the resulting multiplicities gives a monic annihilator of order $S = 46$, which includes the pre-period $n_0 = 15$ the conditions with an offset need. The annihilator was then asked for terms the model had not supplied and reproduced them.

4 The proof

Lemma 1. *Let $A(z) = z^S - \sum_{j < S} \alpha_j z^j$ be the monic annihilator derived above and $q(z) = z^{10} - \sum_i c_i z^{10-i}$ the characteristic polynomial of the conjectured recurrence. The residual $r(n) = a(n) - \sum_i c_i a(n-i)$ is a linear combination of shifts of a , so A annihilates r as well. Since $A(0) \neq 0$ the recurrence it gives runs in both directions, and S consecutive zeros of r force r to vanish at every later index.*

Proof. A annihilates a , hence every shift of a , hence any linear combination of shifts; that is r . A monic recurrence with nonzero constant term determines each term from the S before it and each from the S after it, so a run of S zeros propagates. $\square$

Theorem 2.

$$a(n) = 3a(n-1) - 3a(n-2) + 2a(n-3) - 2a(n-4) + 2a(n-6) - 2a(n-7) + 3a(n-8) - 3a(n-9) + a(n-10)$$

for every $n > 10$.

Proof. The terms $a(0), \dots$ were computed exactly in integer arithmetic from the model, extended by A where the model itself was not evaluated, and $r(n)$ formed for each index. Those integers vanish from $n = 10 + 1$ onwards, and the run exceeds $S + 10$, which by Lemma 1 settles every larger index at once. $\square$

5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 33 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: it is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no model involved, and holds wherever the proved range and the data overlap.

Third, the derived annihilator A was itself tested on terms it had not been given: the model was evaluated beyond the S values that determine it and A reproduced them. A bound that is too small fails this test, which is why it is run.

References

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